Oluwatobi Melvyn Mayungbo

Machine Learning Engineer | Time-Series & Sequence Models | Fraud Detection & Payments

Email | LinkedIn | GitHub

Machine Learning Engineer with a B.Sc. in Statistics, building time-series and sequence models for fraud detection and payments in fintech. I ship production systems in Python (PyTorch, LightGBM, FastAPI, Docker) with calibrated outputs, cost-based decision thresholds and drift monitoring.

EXPERIENCE

SQI College of ICT March – September 2026

Machine Learning Instructor

- Teach machine learning and deep learning to student cohorts (neural networks, sequence models, evaluation) and mentor them through end-to-end projects in Python, PyTorch and scikit-learn.

Resolve Axis January – October 2025

Machine Learning Engineer – Remote

- Built forecasting model on event data to optimize operations.
- Choose decision thresholds using cost-based trade-offs.
- Deployed production systems with monitoring.

NISER May – July 2024

Internal Auditor

- Analyzed financial and operational records to flag irregularities and suspicious patterns, performing manual anomaly detection across transactional datasets.

TECHNICAL PROJECTS

Sequence-Based Transaction Fraud Detection System GitHub | Live App

Tech Stack: Python, PyTorch, LightGBM, ONNX Runtime, FastAPI, Docker, SHAP

- Built GRU, TCN and Transformer fraud models on ~297k simulated transactions; the GRU beat a strong LightGBM baseline on PR-AUC (0.870 vs 0.826, 3 seeds), widening to 0.765 vs 0.615 after adversarial drift.
- Built a real-time scoring API with ONNX Runtime and FastAPI at 7.2 ms p99 latency against a 50 ms budget (ONNX 2.8× faster than PyTorch), with isotonic calibration (ECE ~0.0002) and an audit log.
- Built label-free drift monitoring that caught the attack shift in its first window (alert rate fell to 57% of baseline) while standard score PSI never fired; leakage tests fail the build if any feature reads the future.

Reckon — Payment Reconciliation & Review System GitHub | Live App

Tech Stack: Python, FastAPI, SQLAlchemy, Paystack API, Docker, GitHub Actions, Railway

- Built a Paystack payment-to-invoice matcher: exact rules, then a calibrated logistic model, then human review; on a simulated month of 439 payments (NGN 37.2M) it auto-closed 76.1% at 100% precision.
- Set the auto-close threshold from an 83:1 error-cost ratio, calibrated with Platt scaling (Brier 0.022) and ranked review by money at risk, cutting monthly matching time from 21.9 to 5.2 hours.

Credit Risk Decisioning System GitHub | Live App

Tech Stack: Python, LightGBM, scikit-learn, SHAP, WOE Scorecards, FastAPI, Docker, GCP Cloud Run

- Built a credit decisioning API on a 30,000-application synthetic loan book: LightGBM PD model (out-of-time AUC 0.838, ECE 0.026), cost-optimal cutoffs, SHAP adverse action reason codes and an audit log.
- Showed the cutoff matters more than the model: an F1-chosen cutoff forfeits 28% of achievable profit and a naive 0.5 cutoff loses money outright.

TECHNICAL SKILLS

Machine Learning & Time Series: PyTorch, scikit-learn, LightGBM, Pandas, NumPy, Time-Series Forecasting, Sequence Models (GRU, TCN, Transformers), Quantile Forecasting, Backtesting, Feature Engineering

Fraud, Payments & Risk: Fraud Detection, Credit Risk, Model Calibration, Cost-Based Thresholds, SHAP, Drift Monitoring, Payment Reconciliation

Engineering & MLOps: Python, SQL, FastAPI, Docker, ONNX Runtime, MLflow, Prefect, dbt, GitHub Actions (CI/CD), GCP Cloud Run, Git

EDUCATION

University of Ibadan January 2021 – February 2025
B.Sc. in Statistics — Ibadan

SQI College of ICT July 2025 – July 2026
Professional Diploma in Artificial Intelligence — Ibadan